

ALEXANDER TONG

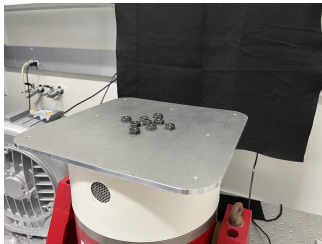
Mechanical engineering | Recovery hardware and supersonic aerodynamics

Boston University, B.S. Mechanical Engineering, Aerospace Concentration | Expected May 2028
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PORTFOLIO AT A GLANCE

Three detailed examples

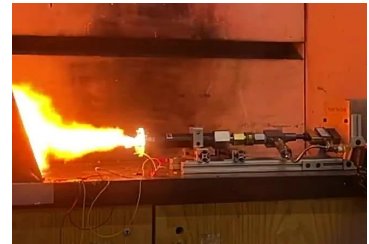
Page 2 / Separation & recovery	18.7 lbm within a 20 lbm allocation . Parachute-load sizing, CO2 hardware and an operational reduced-pressure test chamber.
Page 3 / Aerodynamic fairing	Three concept studies; Ansys CFD baseline converged and under review . Shared aerodynamic analysis, integration and manufacturing trade.
Page 4 / Plummer shaker hardware	Structural analysis, bored-bolt checks and CNC fabrication . Installed plate-and-bolt hardware at 93.9% lower estimated cost.



Plummer: machined adapter installed.



AIAA: assembled Prototype 1 aircraft.



Hybrid nozzle: hotfire test frame.

Other selected work / links to evidence

Plummer Laboratory

CNC-machined shaker adapter; plate-and-bolt hardware estimate reduced from \$1,170 to \$71.11, excluding labor. Designed the matrix and procedure for 48 Instron tests.

AIAA Design-Build-Fly wing

Complete Prototype 1 wing 42.7% lighter than the previous complete wing; build time reduced from three months to one. Led design, fabrication and member training.

Hybrid rocket nozzle

73% mass reduction from my initial design; approximately two-second hotfire. Minimum yield FOS 7.8 in SolidWorks FEA for mechanical loads only; thermal effects excluded. Thrust gain unmeasured.

EK210 line-following car

Designed and programmed the vehicle: six loaded course runs in 70-86 seconds against a 90-second limit. Recorded component cost \$73.94 against a \$200 budget.

Culturon

Executed the company's first full-scale nanoparticle deposition map using microscopy and ImageJ. Designed adhesive-free sample racks; presented prevention concepts.

Manufacturing & mentoring

CNC and manual machining, hot-wire foam cutting and composite fabrication. Taught BURPG members turning and milling through the machinist's-hammer project beginning Fall 2025.

Additional research & coursework

Space Physics & Technology Lab: instrument-data analysis with Python/pySPEDAS and assistance with flight-spare laser calibration.
University of Sydney pulsed plasma thruster: inductor-spool design and coil-winding calculations. [ME360 course project](#) is in progress.

Recovery design within mass budget; hardware tested

Boston University Rocket Propulsion Group | Subsystem Co-Lead | June 2025-present

18.7 / 20 lbm

Subsystem mass / allocation; final BOM
\$1,076

823 / 1,230 lbf

Calculated parachute opening / snatch
loads

352 L

Operational vacuum chamber; reported test
pressure 0.011 atm



Integrated subsystem before CO2 coldflow; full-system separation testing is pending.



Manufactured CO2 cartridge/piercer assembly. Design and fabrication shared by our three-person team.



Door, adapter and seal designed by me; fabricated by my co-lead, who designed the clamps.

My engineering decisions and execution

Co-led architecture and deployment-hardware trades through four reviews. Compared five published load methods to size lines and check hardware; paired flight computers provide separate firing paths.

After integrated black-powder tests exposed charge failures and parachute damage, I sized CO2 piercers using pressure, thick-wall stress and thread-shear calculations; contributed CAD, drawings, machining and testing with a three-person team. Wrote the summer test procedure and helped lead briefings, fire/abort decisions and inspection.

Calculation basis

Recovery lines	Selected nominal 6,000 lbf line rating against the calculated loads, accounting for termination efficiency. Nominal line rating is not assembled splice strength.
CO2 deployment sizing	Calculated 4.66 psi separation requirement for four nylon shear pins plus estimated interface friction. Ideal-gas estimate for the modeled free volume: 10.9 g versus a selected 16 g cartridge.
CO2 piercer housing	Minimum calculated factor of safety (FOS) 2.01 in Lamé hoop stress; internal-thread shear FOS 6.31. Analytical margins, not proof-test results.
Vacuum door / Roark	0.25 in 6061-T6 plate; 12 in unsupported diameter; 14.7 psi differential gives 1,663 lbf total load. Recorded bending stress 16.5 ksi; deflection 0.0848 in; minimum yield FOS 2.12 using 240 MPa yield strength. Uniform pressure; simply supported around the outer circumference. Deflection uses the Roark plate model; stress is a separately entered calculator result.
Legacy Catclaw body	Recorded minimum body FOS 2.01 after my material reduction and piston-retention-slot modification. This tie cutter is separate from the CO2 piercer.

Demonstrated results and remaining verification

Reduced-pressure CO2 component firing is complete. Dynamic splice proof testing reached an estimated 1.5 FOS without observed slip, limited by the setup. Integrated CO2 separation and subscale testing remain pending.

Chamber instrumentation and data analysis by my co-lead. Door stress/deflection are predictions; the recorded stress-based FOS is not an independently reproduced Roark stress result. Selected yield input is not stock certification.

[Recovery architecture, hardware & tests](#) | [CO2 sizing & stress calculations](#) | [Chamber design](#)

Defined the aerodynamic trade; CFD baseline converged

Boston University Rocket Propulsion Group | Project Lead | August 2026-present

63 in × 1.6 in

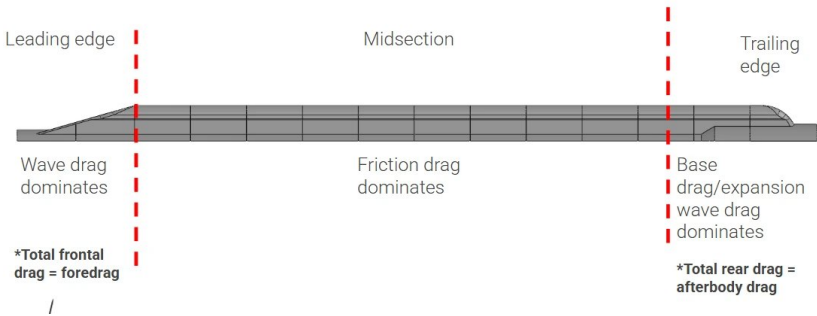
Fairing length / nominal height on an 8 in diameter rocket

Mach 2.0-2.81

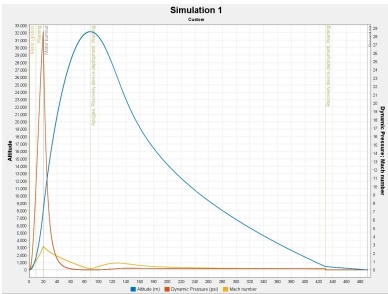
Priority comparison range from the simulated trajectory

3 concepts

Teammate-owned geometries; shared requirements and analysis by me



My CoDR diagram: a first-principles decomposition, not a measured allocation of drag. Pressure and friction act across the whole fairing.



Trajectory used to select high-dynamic-pressure analysis conditions; simulation, not flight data.



Legacy Icarus fairing: the CFD baseline and a backup flight candidate, not a new design by me.

Analysis and design philosophy

Used $D = q S_{ref} C_D$ to prioritize conditions by dynamic pressure as well as Mach. Evaluated leading-edge pressure drag, midsection wetted area and aft pressure recovery together with thermal loads, attachment integrity and manufacturing constraints. Isolated axisymmetric profile rankings cannot establish the performance of a partial fairing attached to the rocket body: junction flow and fin interference require integrated analysis.

Preliminary friction estimate	At Mach 2.81 and recorded $Re = 2.04 \times 10^8$: $C_{D,f} \approx 0.0792$, using 477 in ² wetted area, 7.99 in ² reference area and a 10% preliminary roughness allowance.
Leading-edge surrogate	A 15° full-cone correlation gives $C_p \approx 0.1888$. It is a screening surrogate for the wedge-like legacy front, not the mounted fairing’s total drag coefficient.

Ownership, results and next decision

I developed the common analysis, integration/manufacturing trade and CoDR. Quinn Colleran, Hannah Ruan and Aaron Lin own the three concept geometries. My Icarus simulation in **Ansys CFD is converged and under review**; candidate ranking and adjoint optimization remain pending.

The investigation also led to a nine-profile-family nosecone sweep in RASAero II. A ranking difference from published comparisons prompted further investigation; it does not establish a better flight design.

[Concept credits, hand calculations, sources & nosecone sweep](#) | Hand-calculation reference: S. S. Chin, Missile Configuration Design (1961).

Shaker hardware: analyzed, machined and installed

Mechanical Design Assistant | June 2026-present

0.173 mm

Plate displacement predicted in the design-review FEA case

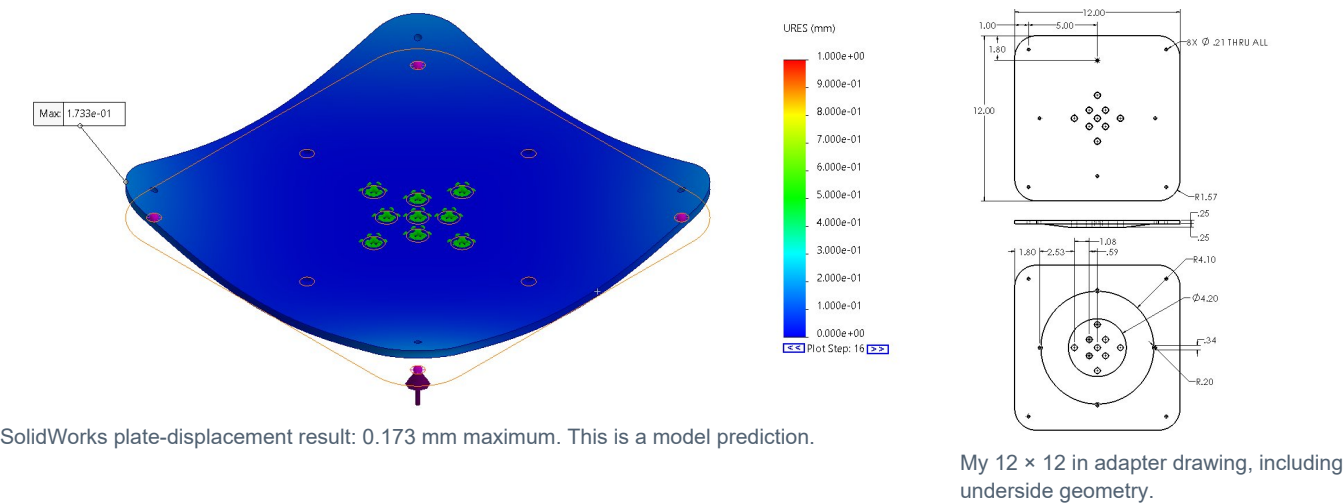
20.1 Hz

Shaker operating frequency; analyzed with a 3 kg fixture load

93.9% lower

Estimated plate-and-bolt hardware cost; labor excluded

The adapter mounts a physical Ising-model fixture to the shaker for controlled vibration tests of bistable buckling elements. It transfers shaker motion into the fixture while maintaining the mounting interface.



Design decision and manufacturing

Replaced an estimated \$1,170 outsourced plate/custom-bolt solution with \$71.11 of in-house hardware. Developed SolidWorks CAD/FEA, drawings and CAM/G-code; machined the 6061-T6 plate on a Haas VF-2 and checked plate dimensions with calipers against the ±0.005 in drawing tolerance.

Drilled 5 mm axial bores in standard medium-carbon-steel M10 bolts for the required shaker interface after unsuccessful stainless-steel machining attempts. The bored bolts and adapter are installed.

Structural calculation basis

Installed joint geometry	M10 × 1.5 with a 5 mm axial bore; modeled remaining tensile area 38.35 mm². Nominal Class 8.8 yield strength 640 MPa; local plate bearing thickness 9.7 mm.
Per-bolt load path	5 kg allowance (2 kg plate + 3 kg fixture) at 11 G, shared by four bolts: 134.9 N external load per bolt. Workbook conservatively adds the full external load to 16.36 kN preload: 16.50 kN combined; joint stiffness sharing is not modeled.
Static FOS by checked mode	Bolt tensile yield: 1.49 under preload plus full external load; preload alone gives 1.50. Plate bearing 2.46; pull-through 4.99; external-thread shear 3.71; internal-thread shear 3.60. These are mode-specific ratios, not an overall dynamic FOS.
Displacement checks	Selected L/240 limit for the 12 in plate span: 1.27 mm; review plate FEA: 0.173 mm. A separate later dynamic fixture model predicts 0.44 mm at 3 kg and 20.1 Hz. These are different load cases.

Preload maintains joint compression; bolt/joint stiffness determines cyclic load sharing. The 1.49 ratio checks static tensile yield, not fatigue at 20.1 Hz. Preload scatter, separation and fatigue are not quantified here. Displacements are predictions; cost covers plate and bolts only.

[Full calculations, drawings, FEA and machining evidence](#) | Sources: my design review and DSRF Calcs; joint interpretation: [NASA TM-106943](#).